🎓 How to Run UniAssist AI

A simple, step-by-step guide. No tech background needed.

What is this?

UniAssist AI is a chat assistant added to your university system. Students can ask it questions like "What are the admission fees?" and it answers using your school's own documents.

This guide shows you how to get it running on your computer.

Part 1: Things to Install First

You only do this once. Think of it like installing apps before you can use them.

1 Python

This runs the "brain" part of the project (the backend).

1. Go to python.org/downloads
2. Click the big yellow "Download Python" button
3. Open the downloaded file
4. **Important:** check the box that says "Add Python to PATH" before clicking Install

2 Node.js

This runs the website part (the frontend) that students will see.

1. Go to nodejs.org
2. Download the version that says **LTS**
3. Open the file and click Next, Next, Install

3 Docker Desktop

This runs the "memory" that stores your documents so the AI can search them.

1. Go to docker.com/products/docker-desktop
2. Download and install it
3. After installing, open the Docker Desktop app once and leave it running in the background

4 An OpenRouter Account

This is what lets the AI actually think and answer questions.

1. Go to openrouter.ai and sign up (free)
2. Click "Keys" and create a new key — copy it somewhere safe
3. Go to openrouter.ai/models and pick a model, copy its name

💡 You will paste the key and model name into a file later. Keep this tab open.

Part 2: Set Up the Project

Do these steps in order, only once.

1 Open the backend folder

Open a terminal (search "cmd" on Windows) and type:

```
cd backend
```

2 Create a safe Python space

This keeps the project's tools separate from everything else on your computer.

```
python -m venv venv
venv\Scripts\activate
```

3 Install the project's tools

```
pip install -r requirements.txt
```

This may take a few minutes. That is normal.

4 Add your secret key

1. Find the file called `.env.example` inside the `backend` folder
2. Make a copy of it and rename the copy to `.env`
3. Open `.env` in Notepad
4. Replace the placeholder text with your real OpenRouter key and model name from Part 1, Step 4
5. Save the file

5 Add your school's documents

Put your `.txt` files (admission rules, fees, policies, etc.) into this folder:

```
backend/data/documents/
```

Part 3: Run the Project

Do these steps every time you want to start the app. Keep all three windows open at the same time.

1 Start the memory (Qdrant)

In a terminal window, type:

```
docker run -p 6333:6333 qdrant/qdrant
```

Leave this window open. It must stay running.

2 Teach the AI your documents

Open a **new** terminal window:

```
cd backend
venv\Scripts\activate
python scripts/rebuild_index.py
```

Do this once, and again any time you add new documents. The first time, it may download a model — this can take a few minutes if your internet is slow.

3 Start the backend (the brain)

In the same window as Step 2:

```
uvicorn app.main:app --reload
```

Leave this window open too.

4 Start the frontend (the website)

Open a **third** terminal window:

```
cd frontend
npm install
npm run dev
```

`npm install` only needs to run the first time.

5 Open it in your browser

The last terminal will show a web address, something like:

```
http://localhost:5173
```

Click it, or copy it into your browser. Then click "**UniAssist AI**" in the left menu and start asking questions!

Quick Checklist (every time you start)

- ☐ Window 1: Docker running (`docker run -p 6333:6333 qdrant/qdrant`)
- ☐ Window 2: Backend running (`uvicorn app.main:app --reload`)
- ☐ Window 3: Frontend running (`npm run dev`)
- ☐ Browser open to the address from Window 3

If Something Goes Wrong

Chat says "knowledge base unavailable"

Docker (Qdrant) is probably not running, or you forgot Step 2 in Part 3 (rebuild_index.py).

Chat says "API key is not set"

Check the `.env` file — make sure your OpenRouter key is pasted in correctly and saved.

AI doesn't know about a new document

You added a new `.txt` file but forgot to run `python scripts/rebuild_index.py` again.

Page won't load in browser

Make sure all three terminal windows are still open and show no red error text.